How-To Library

Informatica

Configuring a Subject Registry Entity in 10.4.1.1

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Abstract

This article describes the properties in a Subject Registry entity configuration file in Informatica® Data Privacy Management 10.4.1.1 and provides general guidelines on how to configure the file. It includes a sample entity file with the property descriptions and reference information that you can use when you create an entity file.

Supported Versions

- Data Privacy Management 10.4.1.1

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Overview

You can create a Subject Registry in Data Privacy Management. A Subject Registry is a repository of all subject data within an organization.

A subject registry search returns information on all data on a subject and the databases within the organization that contain data related to the subject. The Subject Registry includes golden data stores and transaction data stores that contain subject data linked to subjects in golden data stores.

You can define different types or categories of subjects based on the need or reason to collect and store subject data. For example, a Customer category might indicate all data that you collect and store on customers. Employee might indicate all employee data that you store.

Configure entity files to create different types or categories of subjects in the Subject Repository. Each entity file contains a list of data stores that include golden records. The file also contains properties that configure how the Subject Registry search for data works for the category.

When you perform a subject registry search, you can choose the required entity or category that you want to search. For example, choose the `Employee` entity to search for all data related to an employee. Based on how you configure the `Employee` entity file, specific fields appear on the search and results pages.

Sample Entity File

The following code represents a sample entity file:

```
{
    "Version": "1",
    "EntityName": "AllProps",
    "IgnoreNullEmptyValues":"true",
    "GoldenDatastores": [
                {
                "name": "SR_SATS_30008_BANK_HR"
                },
                {
                "name": "SR_SATS_30008_CC_HR"
                },
                {
                "name": "SR_SATS_30008_INSU_HR"
                }
                ],
    "SearchResultLimit":"10",
    "EntityFields": [{
            "FieldName": "EMPLOYEE_ID",
            "FieldLength": "10",
            "DisplayCode": "00",
            "IsPrimaryKey": "true"
        },
        {
            "FieldName": "SR_FULLNAME",
            "FieldLength": "30",
            "DisplayCode": "03",
            "IsSubjectName": "true",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_EMAILADDRESS",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_MOBILENUMBER",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_DATE_OF_BIRTH",
            "FieldLength": "0",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_AGE",
            "FieldLength": "03",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_GENDER",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_SSN",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_MARITALSTATUS",
```

```
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_CITY",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_STATE",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_COUNTRY",
            "FieldLength": "30",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },
        {
            "FieldName": "SR_DESIGNATION",
            "FieldLength": "40",
            "DisplayCode": "02",
            "IsPrimaryKey": "false"
        },{
            "FieldName": "LastUpdatedTime",
            "FieldLength": "00",
            "DisplayCode": "02",
            "IsPrimaryKey": "false",
            "IncrementalScanField":"true"
        },
        {
            "FieldName": "SOURCE",
            "FieldLength": "40",
            "DisplayCode": "00",
            "IsPrimaryKey": "false"
        }
    ],
    "MatchConfigurations": [{
        "MatchConfigurationName": "MatchConfiguration1",
        "UsedIn": "Scan",
        "IndexConfiguration": {
            "FieldName": "SR_FULLNAME",
            "IndexType": "Fuzzy",
            "IndexPurpose": "Person_Name",
            "IndexLevel": "Limited"
        },
        "SearchConfiguration": {
            "FieldName": "SR_FULLNAME",
            "SearchType": "Fuzzy",
            "SearchPurpose": "Person_Name",
            "SearchLevel": "Typical"
        },
        "MatchRules": [{
            "MatchRuleName": "MatchConfig1",
            "MatchPurpose": "Person_Name",
            "MatchLevel": "Typical",
            "MatchThreshold": "75",
            "MatchRulePurposeCode": "3",
            "MatchFields": [{
                    "FieldPurpose": "Person_Name",
                    "FieldName": "SR_FULLNAME",
                    "MatchType": "Fuzzy"
                },
                {
                    "FieldPurpose": "Address_Part2",
                    "FieldName": "SR_CITY",
                    "MatchType": "Exact"
```

```
                }
            ]
        }]
    },
    {
        "MatchConfigurationName": "MatchConfiguration2",
        "UsedIn": "Search",
        "IndexConfiguration": {
            "FieldName": "SR_FULLNAME",
            "IndexType": "Fuzzy",
            "IndexPurpose": "Person_Name",
            "IndexLevel": "Limited"
        },
        "SearchConfiguration": {
            "FieldName": "SR_FULLNAME",
            "SearchType": "Fuzzy",
            "SearchPurpose": "Person_Name",
            "SearchLevel": "Typical"
        },
        "MatchRules": [{
            "MatchRuleName": "MatchConfig2",
            "MatchPurpose": "Person_Name",
            "MatchLevel": "Typical",
            "MatchThreshold": "75",
            "MatchRulePurposeCode": "3",
            "MatchFields": [{
                    "FieldPurpose": "Person_Name",
                    "FieldName": "SR_FULLNAME",
                    "MatchType": "Fuzzy"
                },
                {
                    "FieldPurpose": "Attribute1",
                    "FieldName": "SR_EMAILADDRESS",
                    "MatchType": "Fuzzy"
                },
                {
                    "FieldPurpose": "Attribute2",
                    "FieldName": "SR_SSN",
                    "MatchType": "Exact"
                }
            ]
        }]
    }]
}
```

Entity File Properties

This section describes the main properties in an entity file.

EntityName

Defines the name of the subject type or category. The name of the entity file must be in the following format: `entity_<EntityName>.json`

GoldenDataStores

Defines the list of data stores that contain golden or master records on individual subjects. You must include all data stores that contain golden records on subjects that you want included in the category or subject type. The data stores are used to create a master list of subjects that is stored in the Subject Registry.

Golden data stores must be structured data stores.

The first data store that you add is considered the primary golden data store, and the other data stores are secondary data stores. The scan first scans the primary golden data store and creates the golden data set in HBase. Subjects from the secondary data stores are inserted or updated based on the entity configuration.

SearchResultLimit

Limits the number of results fetched for a subject search. In the sample file, the search returns a maximum of 10 potential subjects for a given search.

IgnoreNullEmptyValues

Determines whether to ignore or reject a null or empty field value when you configure an index field for Exact type matching. If set to true, the scan ignores empty and null value fields during Discover Subject scans and Link Subject scans.

The setting applies to all golden data stores in the entity. Default is true.

Entity Attributes

This section describes the entity attributes in an entity file.

EntityFields

Defines the attributes that you use to uniquely identify a record.

The EntityFields section includes the following attributes:

FieldName

Name of the field. The name must match the data domain linked to the field that you want to retrieve. The name must be the name provided in the SQL.

FieldLength

Size of the field. If the data in the field is more than the specified length, the data is truncated.

DisplayCode

Specifies where the value of the attribute must be displayed:

- 00 - None. Do not display the value.
- 01 - Display the value on the Subject Search Results page.
- 02 - Display the value on the Subject Details page.
- 03 - Display the value in the Subject Search Results page and on the Subject Details page.

IsPrimaryKey

Defines whether the attribute is a primary key attribute. If true, the value must be unique for each individual subject. Only one attribute can be a primary key. Mark this property as true only for a column that contains unique values. For example, ID columns.

Residency

If set to true, this field indicates residency of the subject. You can set the value as true for multiple fields.

IncrementalScanField

Required to configure repeat scans of structured data stores as incremental scans. Enter true or false. Set this property to true for a field that contains a time stamp value. For example, LastModifiedTime. The scan uses the value in this field to determine when the data was last updated, and scans for changes post the date and time mentioned. The field must use the following format: `yyyy-MM-dd HH24:mm:ss`

IsMultiValued

Specifies if the field can include multiple values for a golden record. For example, a customer might have multiple email addresses or phone numbers. Each of these can be part of golden data to be used in matching for transaction data stores.

IsSubjectName

Set to true to indicate that this field contains the subject name. The subject name appears on the Subject Details page and in DSAR reports.

Note: If you do not set the value to true for any field, by default the Search pages display the FullName, FirstName, or LastName field.

"FieldName": "SOURCE" , "FieldLength": " " , "DisplayCode": "00" , "IsPrimaryKey": "false"

Mandatory attribute. Indicates the name of the data store being scanned. The field length should be equal to or greater than the longest data store name included in the scan.

Match Configuration

A match configuration defines the rules to identify and link a subject during a subject registry scan and to search for a subject during a subject search. The configuration includes the search and matching configuration for the entity.

A MatchConfiguration includes the following attribute:

UsedIn

Indicates whether the match configuration is used in subject scans, searches, or both. Use one of the following values:

- Search
- Scan
- Scan, Search
- Search, Scan

A MatchConfiguration includes the following attribute sections:

- IndexConfiguration
- SearchConfiguration
- MatchRules

Index Configuration

Configure index configuration attributes to perform the following tasks:

- Generate the indexes during a golden data store scan.
- Generate ranges during a Link Subjects scan. Applicable for secondary golden data stores and transaction data stores.

An IndexConfiguration includes the following attributes:

FieldName

Defines the name of the entity field on which the index must be generated. You can include multiple field names in an index configuration, but all field names must use the same purpose.

IndexPurpose

Indicates the purpose for which the field is used. A list of purposes is available in IndexConfiguration.

IndexLevel

Defines the extent of fuzziness. Can use values of Limited, Standard, and Extended, with Limited being the least fuzzy and Extended being the most fuzzy. The fuzziness increases as you move from Limited to Standard and the performance decreases.

IndexType

You can set the value to Fuzzy or Exact. Default is Fuzzy.

Search Configuration

Subject scans and subject searches performed on the Data Privacy Management user interface use the search configuration that you configure in the entity file.

The fields that you define must be the same or a subset of the fields that you define in the index configuration.

A SearchConfiguration includes the following attributes:

FieldName

Defines the name of the entity field on which the index must be generated.

SearchType

You can set the value to Default or Fuzzy. Default is Fuzzy.

SearchPurpose

Indicates the purpose for which the field is used. The SearchPurpose must be the same as you define in the IndexConfiguration.

SearchLevel

Defines the accuracy required to determine a match. Values can be Narrow, Typical, Exhaustive, or Extreme, with Narrow being the lowest accuracy and Extreme being the highest accuracy. The search result set decreases as you increase the accuracy.

Match Rules

The MatchRules section defines rules to match a given record against a golden record during a scan. If the match score exceeds the defined threshold, the record is considered a match.

A MatchRule includes the following attributes:

MatchRuleName

Defines the name of the match configuration rule.

MatchPurpose

Specifies the purpose for which this field will be used. A list of match purposes is available in MatchRules.

MatchLevel

Defines the accuracy required to determine a match. Values can be Loose, Typical, or Conservative, with Loose being the lowest accuracy and Conservative being the highest accuracy.

MatchThreshold

Defines the minimum threshold of a match score for a match to be successful. A higher value means more accurate matching.

MatchRulePurposeCode

Code of the MatchPurpose assigned to the match rule. Must be set to 3.

MatchFields

Defines the fields to use for matching. Indicates a minimum set of fields required to consider a match. The greater the number of fields, the higher the accuracy of the match.

FieldPurpose

The purpose assigned to a given field. A list of field purposes is available in the MatchRules.

FieldName

The field name on which you want to generate the index for the entity.

MatchType

Defines the type of matching logic to use. Can be Exact or Fuzzy. Exact indicates that an exact match is required to consider the value a match. Fuzzy indicates that non-exact or approximate matches can be considered as a match.

Subject Registry Rules and Guidelines

When you configure a Subject Registry and the entity files for the Subject Registry, you must analyze the data and the kind of searches that you are likely to perform.

Refer to the rules and guidelines when you configure and set up a Subject Registry.

General Guidelines for Subject Registry Scans

- You cannot include an asterisk (*) in the data because Data Privacy Management uses the asterisk (*) as a delimiter.
- The length of the fields in the `EntityFields` section has a direct relation to the space consumed.
- If you do not want to include a field in searches and scans, set the field length to 0. This improves the performance, and ensures the field is available in DSAR reports.
- If you update an entity, you must run a full scan again. You cannot run an incremental scan after you change an entity configuration.
- The `IndexLevel`, `SearchLevel`, and `MatchLevel` attributes depend on data quality, data volume, and business need. There is no single configuration that works for all scenarios. It is recommended to start with a restrictive level and broaden depending on the requirement.
- Use the full name rather than the first name or last name individually in scans as well as search. Use of the full name results in more accurate matches than use of its individual constituent parts.
- The fields defined in the index configuration are mandatory to search for a subject.
- If you update the index configuration, you must run the subject scan again for the changes to take effect.
- The field that you include in the index configuration must be a field that includes unique values and is present in most or all of the data stores that you expect to be scanned for the entity.
- Indexes included in the match configuration are generated during the golden data store scans.
- The *fieldname* attribute in an `IndexConfiguration` can include multiple comma-separated values. All fields use the same index purpose and index level. Include fields that have the same purpose. A separate index is generated for each field included in the *fieldname* attribute.
- `SearchLevel` cannot be broader than `IndexLevel`. It can be narrower or the same as IndexLevel.
- Indexed fields must not contain null or empty values. If fields contain null or empty values, only the last record with a null or empty value processed is retained in the Subject Registry.
- If you define the IndexType as Exact, IndexLevel and IndexPurpose are ignored.
- Matching is case insensitive for both Fuzzy and Exact match types.

- Exact indexes and matching usually perform better than Fuzzy.
- Use the correct purpose for each field in the Index, Search, and Match configurations. The configuration can have a significant impact on performance and the accuracy of a match.
  For a complete list of purposes and information about purposes that you can use with Data Privacy Management, see Reference Information.
- You can use only the USA Standard Population information with the Subject Registry in Data Privacy Management.

Rules and Guidelines for Match Configurations

- You can add multiple match configurations in an entity. Each additional match configuration increases the load on the scan and has an impact on the scan performance.
- Add separate match configurations for scan and search if you do not include search fields in scans. Ensure that you include scan fields required for search in the search configuration.
- If you create separate search configurations, scan configurations are not considered for search.
- You can create multiple search and scan configurations.
- You can include fields defined in a single search configuration in a search. You cannot include fields from multiple search configurations in a search.
- If you do not mark a match configuration for either scan or search, it is used in both scan and search. If you specify, you must specify at least one configuration for scan and one for search. Scans fail if either configuration is missing.
- Within each match configuration, all fields defined in the `MatchFields` are used for matching a subject.
- Fields defined under *MatchFields* appear in the optional search filters if the MatchType is set to Fuzzy. If the `MatchType` is set to Exact, the options appear in the required filters section.
- If you define multiple match configurations and you search on fields across configurations, results from each configuration are combined and returned.
- You must consider the order of match configurations in the entity file. A scan evaluates match configurations in the order that you define in the configuration file. The order of precedence is from first to last. If a match is found for a match configuration, the remaining match configurations are not processed.
- Add the match configuration with index and match rules that have a higher probability of matching with a subject first in the entity configuration file. This means an index field that is present in most data stores and match rules with a higher chance of matching with a subject must be first in a list of match configurations.
- You might have multiple match configurations that have the same index configuration and search configuration values with different match rules. In such cases, consider creating a single match configuration with multiple match rules.
- An OR condition is used between multiple match rules. The rules have a common index but the match criteria differs in each.
- Use Exact `MatchType` configuration for matching numeric fields for better accuracy and performance during a scan.
- For better accuracy, use a higher match threshold value when you run scans to link subjects to golden records.

Reference Information

You can use the following information when you configure purposes and levels in a match configuration.

Index Purpose

You can use the following purpose values as index purposes:

Purpose	Description
Address_Part1	Includes the part of the address up to but not including the locality "last line." The word order, which represents the position of the address components, must be the normal word order used in your data population.
Person_Name	Used to match the names of people. An application must pass the full person name. The word order represents the position of the first name, middle names, and family names. It must be the normal word order used in your data population. For example, in English-speaking countries the normal word order is: `First Name + Middle Name(s) + Family Name(s)`
Organization_Name	Matches the names of organizations, which can be business names, institution names, department names, agency names, or trading names. The field matches a single name or a compound name such as a legal name and its trading style. The field includes the person name rules.
Generic_Field	Generic_Field can contain either Person Names, Organization Names, or Addresses. It can also contain a combination of these.
Code	The Code field indexes and searches for any value that is in the alphanumeric or numeric format, such as credit card number, telephone number, Vehicle Identification Number, date, and International Standard Book Number. The Code field uses an algorithm that overcomes any errors or variations that are common across the code entities, such as missing digits, different digit orders, and special characters.
Telephone_Number	The Telephone_Number field indexes and searches different variations of the telephone numbers. The Telephone_Number field uses a string-matching algorithm that overcomes any errors or variations, such as incorrect area code or country code, special characters, embedded spaces, different digit orders, use of symbols such as plus (+), transpose errors, concatenated digits, deletion or omission of digits, insertion errors, substitution errors, and transpositions.
Date	The Date field indexes and searches for any types of dates, such as expiry date, date of contract, date of change, and creation date. Use the `Day+Month+Year` format for a date. The algorithm uses the methods and options designed specifically for dates to overcome any errors or variations that are common in the date format.
CreditCard	The CreditCard field indexes and searches credit card numbers, which are generally a 16-digit number. A credit card is electronically associated with a bank, a customer, and the customers' accounts. The CreditCard field uses an algorithm that overcomes any errors or variations, such as different industry identifiers, issuer identifiers, account numbers, special characters, embedded spaces, different digit orders, and missing or extra digits.
VIN	The VIN field indexes and searches the Vehicle Identification Number (VIN). It is a unique code that identifies individual motor vehicles, towed vehicles, motorcycles, scooters, or mopeds. The VIN field uses an algorithm that overcomes any errors or variations, such as vehicle manufacturer codes, special characters, embedded spaces, different letter or digit orders, transpose errors, concatenated digits, transpositions, and sequence variations.

Purpose	Description
ISBN10 and ISBN 13	The ISBN10 and ISBN13 fields index and search the International Standard Book Number (ISBN). The International Standard Book Number is a unique code that identifies the commercial books. Use the ISBN10 field to search for a 10-digit number, and use the ISBN13 field to search for a 13-digit number.
Geocode	Use the Geocode field to index and search records based on latitude and longitude geographic coordinates with an optional elevation. Specify the elevation value with its unit after the geographic coordinates. The supported units for elevation are ft (feet), km (kilometer), and m (meter). Negative elevation represents depths below sea level.
Company_Name	Matches the names of organizations, which can be business names, institution names, department names, agency names, or trading names. The Company_Name field matches a single name or a compound name such as a legal name and its trading style. The Company_Name does not include the person name rules.

Using the USA Standard Population, you can index a user database on Person Names, Organization Names, and Addresses using one of three index levels:

Level	Description
Standard	Standard is the recommended Key Level for typical applications. It overcomes most variations in word order, missing words and extra words. It also maximizes the likelihood of finding candidates in cases of severe spelling error in multi-word names. Standard is the default if no Key Level is specified.
Extended	For high-risk or critical search applications, SSA-NAME3 can generate "Extended" Keys. Extended Keys extend Standard Keys by adding more keys based on token concatenation. The designer/developer should be aware that the use of Extended Keys will increase disk space requirements and result in larger candidate sets at search time. However, the intended use of Extended Keys is to improve reliability by finding matches regardless of word order variation and concatenation.
Limited	If disk space is limited, SSA-NAME3 can generate "Limited" Keys. Limited Keys are a subset of Standard Keys. The designer/developer should be aware that the use of Limited Keys, while saving on disk space, might also reduce search reliability.

Search Configuration

You can use the following purpose values as search purposes:

Purpose	Description
Address_Part1	Includes the part of the address up to but not including the locality "last line." The word order, which represents the position of the address components, must be the normal word order used in your data population.
Person_Name	Used to match the names of people. An application must pass the full person name. The word order represents the position of the first name, middle names, and family names. It must be the normal word order used in your data population. For example, in English-speaking countries the normal word order is: `First Name + Middle Name(s) + Family Name(s)`
Organization_Name	Matches the names of organizations, which can be business names, institution names, department names, agency names, or trading names. The field matches a single name or a compound name such as a legal name and its trading style. The field includes the person name rules.

Purpose	Description
Generic_Field	Generic_Field can contain either Person Names, Organization Names, or Addresses. It can also contain a combination of these.
Code	The Code field indexes and searches for any value that is in the alphanumeric or numeric format, such as credit card number, telephone number, Vehicle Identification Number, date, and International Standard Book Number. The Code field uses an algorithm that overcomes any errors or variations that are common across the code entities, such as missing digits, different digit orders, and special characters.
Telephone_Number	The Telephone_Number field indexes and searches different variations of the telephone numbers. The Telephone_Number field uses a string-matching algorithm that overcomes any errors or variations, such as incorrect area code or country code, special characters, embedded spaces, different digit orders, use of symbols such as plus (+), transpose errors, concatenated digits, deletion or omission of digits, insertion errors, substitution errors, and transpositions.
Date	The Date field indexes and searches for any types of dates, such as expiry date, date of contract, date of change, and creation date. Use the `Day+Month+Year` format for a date. The algorithm uses the methods and options designed specifically for dates to overcome any errors or variations that are common in the date format.
CreditCard	The CreditCard field indexes and searches credit card numbers, which are generally a 16-digit number. A credit card is electronically associated with a bank, a customer, and the customers' accounts. The CreditCard field uses an algorithm that overcomes any errors or variations, such as different industry identifiers, issuer identifiers, account numbers, special characters, embedded spaces, different digit orders, and missing or extra digits.
VIN	The VIN field indexes and searches the Vehicle Identification Number (VIN). It is a unique code that identifies individual motor vehicles, towed vehicles, motorcycles, scooters, or mopeds. The VIN field uses an algorithm that overcomes any errors or variations, such as vehicle manufacturer codes, special characters, embedded spaces, different letter or digit orders, transpose errors, concatenated digits, transpositions, and sequence variations.
ISBN10 and ISBN 13	The ISBN10 and ISBN13 fields index and search the International Standard Book Number (ISBN). The International Standard Book Number is a unique code that identifies the commercial books. Use the ISBN10 field to search for a 10-digit number, and use the ISBN13 field to search for a 13-digit number.
Geocode	Use the Geocode field to search records based on latitude and longitude geographic coordinates with an optional elevation. Specify the elevation value with its unit after the geographic coordinates. The supported units for elevation are ft (feet), km (kilometer), and m (meter). Negative elevation represents depths below sea level.
Company_Name	Matches the names of organizations, which can be business names, institution names, department names, agency names, or trading names. The Company_Name field matches a single name or a compound name such as a legal name and its trading style. The Company_Name does not include the person name rules.

Using the USA Standard Population, you can configure an application to search on Person Names, Organization Names, and Addresses using one of four search levels:

Level	Description
Typical	A Typical search level for most applications will provide a practical balance between quality and response time. Use in typical online or batch transaction searches. If you do not specify a Search Level, Typical is the default. For Person_Name searches, it is designed to find common but not extreme error and variation including cases where initials are present instead of full given names and where the initial of a name has changed due to the internal rules applied. For Organization_Name searches, it is designed to find common but not extreme error and variation including instances of word concatenation. For Address_Part1 searches, it is designed to find common but not extreme error and variation.
Exhaustive	An Exhaustive search level is provided for applications that have an increased risk associated with missing a match, where data quality is a concern or where data volumes are low enough to make it the default search. It increases the number of candidates returned and consequently response times might be extended. An Exhaustive search will occasionally find matches that a Typical search misses; however, these will generally be where there is more extreme error and variation. For Person_Name searches, it is designed to find more error and variation than a Typical search, especially where there is extreme spelling error in the family or middle names. For Organization_Name searches, it is designed to find more error and variation than a Typical search, especially where there is extreme spelling error in the major word or trailing words. For Address_Part1 searches, it is designed to find more error and variation than a Typical search, especially where there are more cases of missing words, extra words or sequence differences.
Narrow	A Narrow search level compromises on completeness of search in favor of faster and more direct answers. It can be an option in search applications that do not have a high risk associated with missing a match, require very tight levels of matching, or where data volumes are extreme and response time is a critical factor. For Person_Name searches, it is designed to find the very common error and variation including cases where initials are present instead of full given names. For Organization_Name searches, it is designed to find the very common error and variation and primarily where the words are in a stable order. For Address_Part1 searches, it is designed to find the very common error and variation and primarily where the tokens are in a stable order.
Extreme	An Extreme search level uses every possibility to discover a candidate match. As a result, response times might be extended. It is provided for applications that have a critical need to find a match if one is present in the database, despite the error and variation. An Extreme search might only occasionally find matches that an Exhaustive search misses. However, because the risk is very high, every possible match is deemed important. The types of candidates returned for all Field types is the same when using an Extreme search. Extreme spelling error is picked up in names or addresses with two or more words or tokens.

Match Rules

Each match purpose supports a combination of mandatory and optional field purposes, and each field purpose is weighted according to its influence in the match decision. See the Identity Resolution Best Practices Guide for more information on purposes and weights.

You can use the following `FieldPurpose` values with each corresponding `MatchPurpose`:

Match Purpose	Field Purpose	Description
Wide Contact	Choose from the following field purposes: - Organization_Name - Company_Name - Person_Name - Attribute1 - Attribute2 - Code	The Wide Contact purpose loosely identifies a contact within an organization, without regard to the location.
Contact	Choose from the following field purposes: - Organization_Name - Company_Name - Person_Name - Address_Part1 - Address_Part2 - Postal_Area - Telephone_Number - Id - Date - Attribute1 - Attribute2 - Code - Geocode	The Contact purpose identifies a contact within an organization at a specific location.
Individual	Choose from the following field purposes: - Id - Date - Person_Name - Attribute1 - Attribute2 - Code	The Individual purpose identifies a specific individual by name and with either the same ID number or Date of Birth attributes.
Resident	Choose from the following field purposes: - Address_Part1 - Person_Name - Address_Part2 - Postal_Area - Telephone_Number - Id - Date - Attribute1 - Attribute2 - Code - Geocode	The Resident purpose identifies a person at an address.
Address	Choose from the following field purposes: - Address_Part1 - Address_Part2 - Postal_Area - Telephone_Number - Id - Date - Attribute1 - Attribute2 - Code - Geocode	The Address purpose Identifies an address match. The address might be postal, residential, delivery, descriptive, formal, or informal.

Match Purpose	Field Purpose	Description
Organization	Choose from the following field purposes: - Address_Part1 - Organization_Name - Company_Name - Address_Part2 - Postal_Area - Telephone_Number - Id - Attribute1 - Attribute2 - Code - Geocode	The Organization purpose matches organizations primarily by name. It is targeted at online searches when a name-only lookup is required and a human is available to make the choice. Matching in batch requires other attributes in addition to name to make match decisions.
Household	Choose from the following field purposes: - Person_Name - Address_Part1 - Address_Part2 - Postal_Area - Telephone_Number - Attribute1 - Attribute2 - Code - Geocode	The Household purpose identifies matches where individuals with the same or similar family names share the same address.
Person_Name	Choose from the following field purposes: - Address_Part1 - Date - Person_Name - Address_Part2 - Postal_Area - Telephone_Number - Id - Attribute1 - Attribute2 - Code - Geocode	The Person_Name purpose identifies a person by name. It is targeted at online searches when a name-only lookup is required and a human is available to make the choice. Matching in batch requires other attributes in addition to name to make match decisions.
Fields	Choose from the following field purposes: - Person_Name - Organization_Name - Company_Name - Address_Part1 - Address_Part2 - Postal_Area - Telephone_Number - Id - Date - CreditCard - VIN - ISBN10 - ISBN13 - Attribute1 - Attribute2 - Code - Generic_Field - Geocode	The Fields purpose is provided for general nonspecific use. It is designed in such a way that there are no required fields. All field types are available as optional input fields.

Match Purpose	Field Purpose	Description
Corp_Entity	Choose from the following field purposes: - Address_Part1 - Address_Part2 - Postal_Area - Organization_Name - Company_Name - Telephone_Number - Id - Date - Attribute1 - Attribute2 - Code - Geocode	The Corp_Entity purpose identifies an organization based on its legal corporate name, including the legal endings such as INC and LTD. Use the Corp_Entity purpose in applications that honor differences between names such as ABC TRADING INC and ABC TRADING LTD.
Family	Choose from the following field purposes: - Person_Name - Address_Part1 - Telephone_Number - Address_Part2 - Postal_Area - Attribute1 - Attribute2 - Code - Geocode	The Family purpose identifies matches where individuals with the same or similar family names share the same address or the same telephone number.
Wide_Household	Choose from the following field purposes: - Person_Name - Address_Part1 - Telephone_Number - Address_Part2 - Postal_Area - Attribute1 - Attribute2 - Code - Geocode	The Wide_Household purpose identifies matches where the same address is shared by individuals with the same family name or with the same telephone number.
CC_Owner	Choose from the following field purposes: - Person_Name - CreditCard - Telephone_Number - Postal_Area - Address_Part1 - Address_Part2 - Date - Geocode	The CC_Owner purpose identifies and matches a credit card owner and the address.
CC_Issuer	Choose from the following field purposes: - CreditCard - Organization_Name - Company_Name	The CC_Issuer purpose identifies and matches the organization that issues credit cards to customers.

Match Purpose	Field Purpose	Description
VIN_Owner	Choose from the following field purposes: - VIN - Person_Name - Address_Part1 - Address_Part2 - Organization_Name - Company_Name - Postal_Area - Attribute1 - Attribute2 - Date - Code - Geocode	The VIN_Owner purpose identifies and matches the owner of a vehicle.
VIN_Manufacturer	Choose from the following field purposes: - VIN - Organization_Name - Company_Name - Address_Part1 - Address_Part2 - Postal_Area - Geocode	The VIN_Manufacturer purpose identifies and matches the manufacturer of a vehicle for a specified vehicle number.
AuthorISBN	Choose from the following field purposes: - Person_Name - ISBN10 - ISBN13	The AuthorISBN purpose identifies and matches an author for a specified book with an International Standard Book Number.
PublisherISBN	Choose from the following field purposes: - Organization_Name - Company_Name - ISBN10 - ISBN13 - Address_Part1 - Address_Part2 - Postal_Area - Geocode	The PublisherISBN purpose identifies and matches the publishing organization for a specified book with an International Standard Book Number (ISBN). The ISBN can be a 10-digit number or a 13-digit number.
Geocode	Geocode	Use the Geocode field to match records based on latitude and longitude geographic coordinates with an optional elevation. Specify the elevation value with its unit after the geographic coordinates. The supported units for elevation are ft (feet), km (kilometer), and m (meter). Negative elevation represents depths below sea level. The default key length of the Geocode field is 8 bytes. You can change the key length to 5 bytes by using the KEY_SIZE control.

Using the USA Standard Population, you can configure an application to match on a defined match purpose using one of the following three match levels:

Level	Description
Typical	A Typical match level for most applications delivers "reasonable" matches. Use in typical online or batch transaction searches. It is the default if no match level is specified.
Conservative	A Conservative match level for most applications delivers "close" matches. It is generally used in batch systems where accuracy of match is paramount.
Loose	A Loose match level for most applications delivers matches with a higher degree of variation than Typical. It is generally used in systems where the risk of missing a match is high and manual review is available.

Author

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